Remote Workforce & Employee Well-being Analytics

A Power BI project by Swaraj Madavi

Project Overview

This project analyses a dataset of 5,000 employees across 6 regions, 7 industries and 7 job roles to understand how remote, hybrid and on-site work relate to productivity, stress, satisfaction and mental health. The deliverable is an interactive, nine-page Power BI report: six analytical dashboards, three drill-through pages, synced slicers, dynamic titles, a custom tooltip, hierarchies and bookmarks.

Tools used: Excel, Power BI Desktop, Power Query, DAX

Headline KPIs

KPI	Value
Total Employees	5,000
Average Stress Score	2.01
Average Productivity Score	1.97
Average Satisfaction Score	2.00
Average Sleep Quality	2.00
Average Work-Life Balance	2.98

Key Takeaways

- The workforce is split almost evenly across remote (34.28%), hybrid (32.98%) and on-site (32.74%) work.
- Work location has no effect on average satisfaction, but remote employees report the strongest company support (3.04).
- Access to mental health resources does not appear to improve satisfaction in this dataset.

- Europe is the top priority region for wellness initiatives, with Asia as a secondary focus.

Phase 1: Data Preparation

All cleaning and feature engineering was done in Power Query.

1. Reviewed the source Excel/CSV file for correctness.
2. Loaded the file into Power BI and opened **Transform Data**.
3. Verified the data type of every column; no mismatches were found.
4. Checked Employee ID for missing and duplicate values; none were found.
5. Used column statistics to confirm there were no empty rows.
6. Shortened long column names without changing their meaning and removed hyphens for a cleaner, more professional look.
7. Created three derived columns using **Add Conditional Column**:

New Column	Source	Categories
Age Groups	Age	Under 25, 25-34, 35-44, 45-54, 55+
Experience Groups	Experience (years)	Fresher (up to 1), Junior (up to 3), Mid Level (up to 7), Senior (up to 12), Expert (above 12)
Working Hour Category	Weekly Hours	Less than 35, 35 to 45, 45 to 55, More than 55

The changes were then applied with **Close & Apply**.

Phase 2: Data Modelling

The table was renamed **EMPLOYEE DATA**, and its fields were grouped into display folders in Model view so the field list stays readable.

Display Folder	Fields
Employee Attributes	Age, Age Groups, Employee ID, Experience, Experience Groups, Gender, Industry, Job Role, Region

Display Folder	Fields
Work Characteristics	Work Location, Weekly Hours, Working Hour Category, Virtual Meeting
Mental Health Indicators	Mental Health Condition, Mental Health Resources, Stress Level, Social Isolation, Sleep Quality, Physical Activity
Productivity Indicators	Productivity Change, Work Life Balance, Remote Work Satisfaction, Company Support

Phase 3: DAX Calculations

Measures are kept separate from source columns in dedicated display folders: Basic Measures, Employee Distribution Measures, Productivity Measures, Mental Health Measures, Satisfaction Measures and Numerical Measures.

Basic Measures

Total Employees, Average Age, Average Experience, Average Working Hours, Average Virtual Meetings, Average Work-Life Balance, Average Remote Work Satisfaction, Average Sleep Quality, Average Social Isolation and Average Company Support.

```
Total Employees = DISTINCTCOUNT('EMPLOYEE DATA'[Employee ID])
Average Age = AVERAGE('EMPLOYEE DATA'[Age])
Average Working Hours = AVERAGE('EMPLOYEE DATA'[Weekly Hours])
```

Employee Distribution Measures

Headcount measures that filter Total Employees by one attribute value: gender (2), work location (3), region (6), job role (7) and industry (7).

```
Remote Employees =
CALCULATE(
    [Total Employees],
    'EMPLOYEE DATA'[Work Location] = "Remote"
)
```

Productivity Measures

```
Increased Productivity Employees =
CALCULATE(
    [Total Employees],
    'EMPLOYEE DATA'[Productivity Change] = "Increase"
)

Productivity Improvement Percentage =
DIVIDE(
    [Increased Productivity Employees],
    [Total Employees],
    0
) * 100
```

Similar measures count employees with decreased productivity and with no change.

Mental Health Measures

High, Medium and Low Stress Employees; Employees with Mental Health Conditions; Employees with Mental Health Resources; and Access Percentage.

```
Access Percentage =
DIVIDE(
    [Employees with Mental Health Resources],
    [Total Employees],
    0
)
```

Satisfaction Measures

Average Satisfaction, Average Work-Life Balance, Average Company Support, Average Sleep Quality and Average Physical Activity.

Phase 4: Dashboards

Handling Categorical Fields

While building the first KPIs, I found that several fields (Productivity Change, Stress Level and others) are stored as text such as Increase, Decrease and No Change, so they cannot be averaged directly. To solve this, I created numeric score columns for productivity,

satisfaction, sleep quality and stress, and placed the related measures in a separate **Numerical Measures** display folder.

Report Pages

Page	Business Questions	Key Visuals
1. Executive Overview	How is the workforce distributed?	KPI cards; employees by region (map), work location (donut), gender (donut), industry and job role (bars)
2. Productivity Analytics	Which location is most productive? Do experience and meetings affect productivity?	Productivity by work location, industry, job role, experience group, hours worked and virtual meetings
3. Employee Well-being	Who reports the highest stress? Do work-life balance and sleep affect stress?	Stress distribution; stress by region and job role; work-life balance by location; sleep quality and physical activity by stress level
4. Mental Health Analytics	Who has access to resources, and does access help?	Access donut; satisfaction and stress by access; condition by region and job role
5. Remote Work Analysis	Which location offers the best balance, happiness and support?	Satisfaction, company support, work-life balance, productivity and stress by work location
6. HR Insights	Which regions and roles need attention?	Regions ranked by stress, satisfaction, productivity and company support; roles ranked by stress, work-life balance, isolation and sleep
Drill-through: Region, Job Role Details, Industry Details	What drives the numbers for one segment?	KPI cards for the selected region, role or industry

Interactivity and Design

- **Slicers:** Region, Industry, Work Location, Gender, Job Role and Stress Level, synced across all pages.
- **Dynamic title:** A DAX measure reads all six slicer selections, so each page title reflects the active filters.
- **Navigation:** Buttons link all report pages.
- **Tooltip:** A custom report-page tooltip for Employee Well-being.
- **Hierarchies:** Region, Industry and Job Role.
- **Bookmarks:** One for each task in the project brief.
- **Theme:** Dark theme applied after all functionality was tested, including slicers, cross-filtering and drill-through.

Business Questions and Findings

#	Question	Finding
1	What share of employees work remotely, hybrid and on-site?	Remote 34.28%, Hybrid 32.98%, On-site 32.74%.
2	Which work location has the highest satisfaction?	None. All three locations have the same average satisfaction score.
3	Which job roles experience the highest stress?	Project Manager is the highest, and is also the largest role group (738 employees, 14.76% of the workforce).
4	Does higher work-life balance correlate with higher productivity?	No clear positive correlation; higher balance does not consistently mean higher productivity.
5	Which regions have the greatest social isolation?	Oceania has the highest average social isolation score ([insert score]).
6	Does access to mental health resources improve satisfaction?	No. Employees without access report slightly higher satisfaction (2.01) than those with access (1.99).

#	Question	Finding
7	How do working hours affect stress and work-life balance?	Employees working more than 55 hours show the highest stress (2.05); those working 35 to 45 hours show the lowest work-life balance (2.9).
8	Does sleep quality influence productivity?	Barely. Sleep Quality 2 and 3 average 1.97 versus 1.96 for Sleep Quality 1.
9	Which industries report the highest stress?	Education, with an average stress score of 2.05.
10	Which experience group is most satisfied?	Fresher, with an average satisfaction score of 2.05.
11	Which work location receives the strongest company support?	Remote, with an average score of 3.04.
12	Which regions should HR prioritise for wellness initiatives?	Europe: highest average stress (2.04), relatively low satisfaction (2.02) and the lowest company support (2.96). Asia is a secondary focus because it has the lowest satisfaction (1.97).

Recommendations

1. **Prioritise Europe** for wellness programmes, combining stress support with stronger company backing.
2. **Review Asia's satisfaction drivers**, since it has the lowest satisfaction score.
3. **Monitor long working hours.** Stress is highest above 55 hours per week.
4. **Evaluate the effectiveness of mental health resources.** Access alone did not lift satisfaction, so look at how the resources are used and their quality.
5. **Support Project Managers and Education-sector staff,** the highest-stress role and industry.

Limitations

- Differences between groups are small (often 0.01 to 0.05 on a roughly 1 to 3 scale), so findings are indicative rather than conclusive.

- Scores for stress, productivity and satisfaction come from categorical fields mapped to numbers.
- The analysis is descriptive and shows association, not causation.